Two attack-graph claims disproved by exact computation (2026-11-02)

Verdict. - **Target A, the reverse repair of Nagao's Lemma 4** (`d'_F ≤ d_F`, field equations outside): REFUTED as stated. The repair survives only with an added hypothesis. - **Target B, "last fall degree ≤ degree of regularity" with the Bardet–Faugère–Salvy `d_reg`:** REFUTED.

Both results are exact. Each witness has a hand proof, an independent native or exact computation, and a core-Lean `decide +kernel` module that has passed the repository proof gate (run 41396053760).

No speedup, cost or cross-method claim is made, so there is no matched-rho row.

Protocol: PROTOCOL.md, frozen before computing.

Target A: the candidate repair `d'_F ≤ d_F` (outside)

After Nagao's Lemma 4 (`d_F ≤ d'_F`) was refuted (papers/ic-negative-results, S5), the corpus recorded the reverse inequality, with the field equations outside the generating set, as a candidate repair (`GAP-LEMMA4-REVERSE`). The prior census had no counterexample, but it contained only multilinear generators.

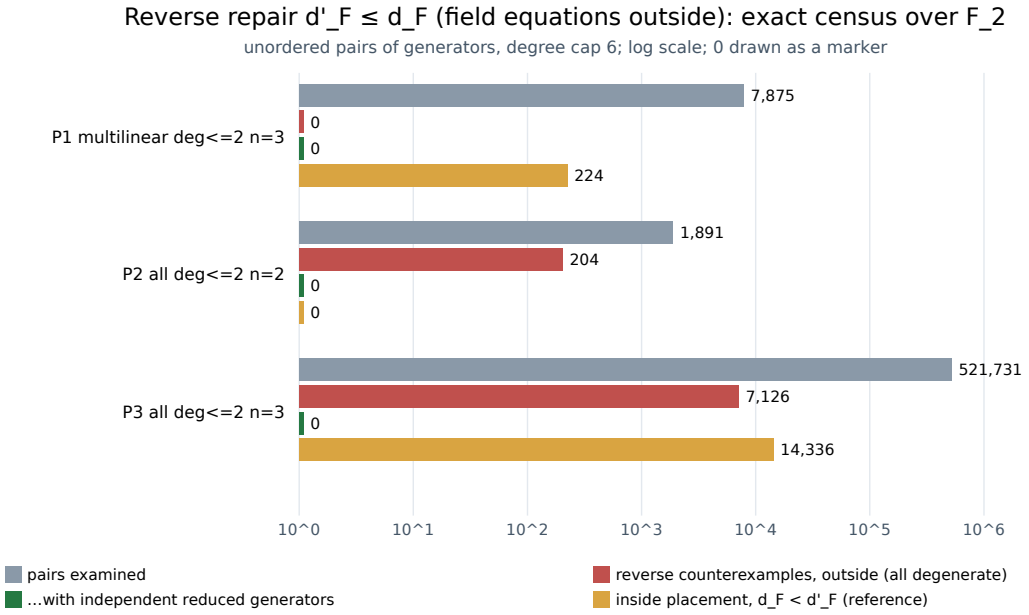
Witness R. Over `F_2`, take `f_1 = X` and `f_2 = X^2 + X`.

- **True first fall degree `d_F = 2`.**
  - `X·f_1 + 1·f_2 = X^2 + X^2 + X = X`. Both products have degree 2 and the sum has degree 1 and is nonzero, so this is a fall at degree 2.
  - At degree ≤ 1 the only admissible multiples are constants times `X`, and the sum of those has the same degree as its largest product, so there is no fall below 2.
- **No fake fall at any degree (`d'_F = ∞`).**
  - Modulo `X^2 + X`, `f_2` reduces to `0`, so every product `g·f_2` reduces to `0`.
  - The reduced sum is then the single product `red(g·X)`, whose degree equals the maximum, so it can never drop.

Hence `d'_F = ∞ > 2 = d_F`.

Census. The instrument is `reverse_census.rs`. Its first 384 lines are a byte-for-byte copy of the fall computations in `blindspots_19_checks.rs` (sha256 `751277a8…`); the new driver adds the census and the degeneracy classification. Unordered pairs, degree cap 6:

population	pairs	d_F > d'_F out / in	reverse counterexamples, outside	of which reduced generators independent	inside, d_F < d'_F
P1 multilinear, deg ≤ 2, n = 3 (control)	7,875	6,138 / 4,830	0	0	224
P2 all, deg ≤ 2, n = 2	1,891	1,446 / 480	204	**0**	0
P3 all, deg ≤ 2, n = 3	521,731	500,108 / 312,718	7,126	**0**	14,336



census

- **P1 reproduces the prior census exactly** (the control passes). It has no “no fake fall” pairs, so the prior “0 outside” was not missing such cases.
- **Every outside counterexample is degenerate.** The two generators become linearly dependent after reduction: one reduces to `0`, or both reduce to the same polynomial. The reduced system then has at most one effective generator and cannot fall.
- **Every counterexample in the census is of the no-fake-fall kind** (`d'_F = ∞`). None has both degrees finite with `d'_F > d_F`.

What survives. `d'_F ≤ d_F` (outside) for systems whose reduced generators are linearly independent: 0 counterexamples among the 531,497 pairs of P1–P3 (P2’s polynomials also occur in P3). This is the new open form of `GAP-LEMMA4-REVERSE`.

Deviation. P4 (multilinear, degree ≤ 3, n = 4) was not run. Its 2^15 polynomials give about 5·10^8 pairs, which is not feasible as an exhaustive census here. The next census should be a seeded sample or three-generator systems (`RT-LEMMA4-REVERSE`).

Target B: last fall degree versus BFS degree of regularity

HKYY state that the last fall degree is bounded by “the degree of regularity”. In their paper that means a solving degree, and the statement holds (`CLAIM-GB-LFD-LE-DREG`). Later introductions repeat the sentence, where the reader takes `d_reg` in the BFS sense. The corpus recorded Salizzoni’s Example 3.9 as the counterexample, but its last-fall value was inferred, not computed.

Witness. `F = {x^2 + y, y^2 + x, xy}` (Salizzoni Ex. 3.9, k = 2).

- `d_reg = 2`:
  - the top-degree parts `x^2, y^2, xy` span all of `R_2`;
  - no generator has degree ≤ 1, so `(F^top)_1 = 0 ≠ R_1`.
- `x, y ∉ V_{F,2}`: all generators have degree 2, so `V_{F,2}` is their span with constant coefficients, and no such combination equals `x` or `y`.
- `x, y ∈ V_{F,3}`, with every product of degree ≤ 3:
  - `x = (y^2 + x) + y(x^2 + y) + x(xy)`;
  - `y = (x^2 + y) + x(y^2 + x) + y(xy)`.

  These identities hold over `F_2`; over other fields the computation finds the corresponding combinations.
- `V_{F,d} = (x, y) ∩ R_{≤d}` for `d = 3..6`: the dimension check passes in every field tried.
- Hence `d_x = d_y = 3`. Caminata–Gorla Theorem 2.9, with the reduced Gröbner basis `{x, y}`, gives last fall degree `3 > 2 = d_reg`.

`salizzoni_ex39.py` runs these checks with exact linear algebra over `F_2`, `F_3`, `F_5` and `Q`; all PASS (`results/salizzoni_ex39.txt`).

Formal status

module	statements	state
papers/ic-negative-results/lean/IcNegative/Lemma4Reverse.lean	reverse_fails_witnessR: the true fall at 2, no fall below 2, no fake fall at any degree	**kernel-checked**: proof gate run 41396053760, target PASS, 32 theorems (8 new), 0 trusted
research/disprove_easy_targets_20261102/lean/SalizzoniEx39.lean	top parts and d_reg, x, y ∉ V_{F,2}, the two V_{F,3} certificates (over F_2)	**kernel-checked**: proof gate run 41396053760, PASS, 4 theorems, 0 trusted

The modules were written without a local Lean toolchain; a Python mirror of the same functions agreed with every statement before the gate ran.

The same gate runs (41390917050, 41396053760: 97 targets, 0 failures) also cover research/jksst_xedni_audit_20261101/lean (294 theorems), so the four JKSST counterexample claims in the corpus are now recorded as kernel-checked.

What this does not say

- Nothing about Nagao's Proposition 5, or about any cost estimate. Neither refuted claim is a premise of a repository cost row; the attack graph shows no claim resting on either.
- The surviving form of the reverse repair is supported only for pairs, degree ≤ 2 and n ≤ 3.
- Target B refutes a misreading, not HKYY's statement. It matters wherever a repository file uses "degree of regularity" without saying which one (RT-GB-DREG-READING-AUDIT).

Reproduce

```
D=research/disprove_easy_targets_20261102
rustc --edition 2021 -C opt-level=3 -o /tmp/rc $D/reverse_census.rs
/tmp/rc p1; /tmp/rc p2; /tmp/rc p3          # ~0.4 s, ~0.04 s, ~30 s on the session container
python3 -I $D/salizzoni_ex39.py
python3 -I $D/plot_census.py                # results/census.svg from results/p*.txt
```